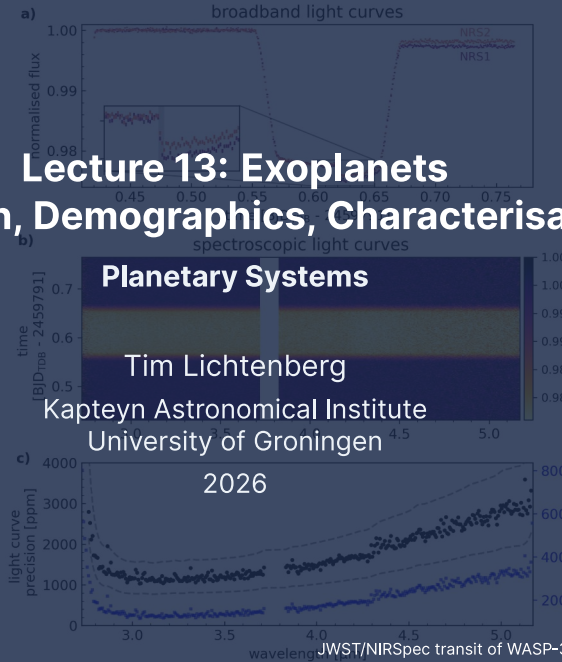


# Lecture 13: Exoplanets

## Detection, Demographics, Characterisation



# Learning Objectives

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By the end of this lecture, you will be able to:

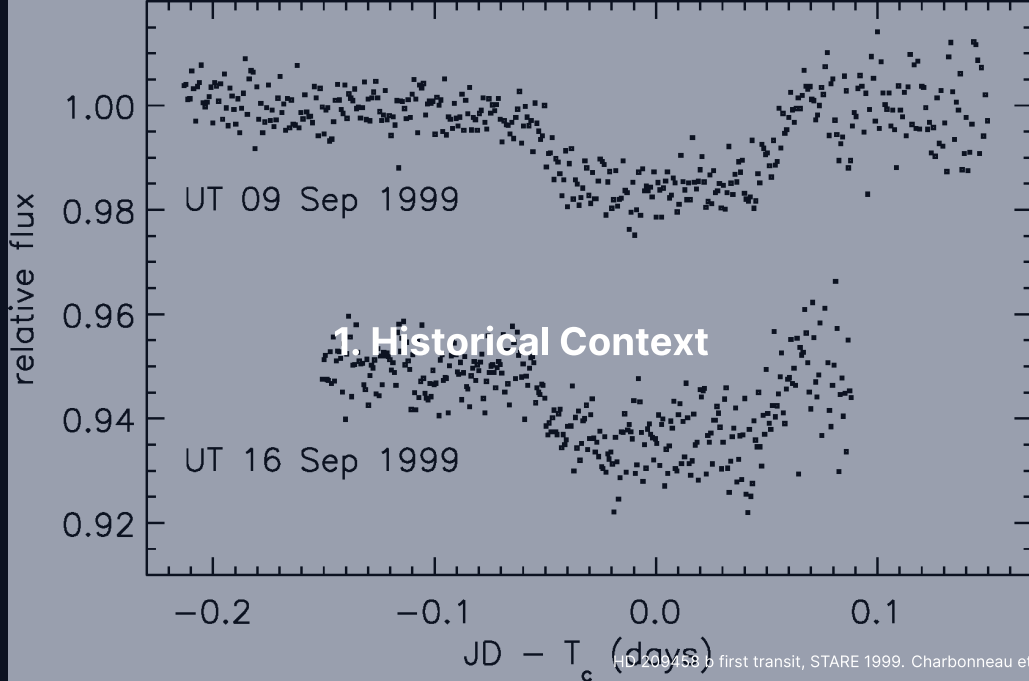
- Describe the main exoplanet detection methods and their biases.
- Interpret the period-radius diagram and the radius valley.
- Apply transit + radial-velocity geometry to derive mass, radius, and bulk density.
- Evaluate JWST-era atmospheric results and biosignature claims.

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

## Recap: Where We Are in the Course

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- L9-L11 walked through every major solar system planet.
- L12: small bodies as formation fossils.
- Today: *turn the question outward*. Are these patterns universal?
- L14: synthesis. What constitutes a convincing detection of life?



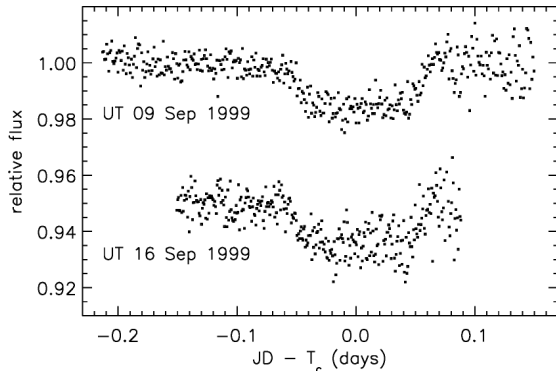
HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

## From Zero to Six Thousand in Three Decades

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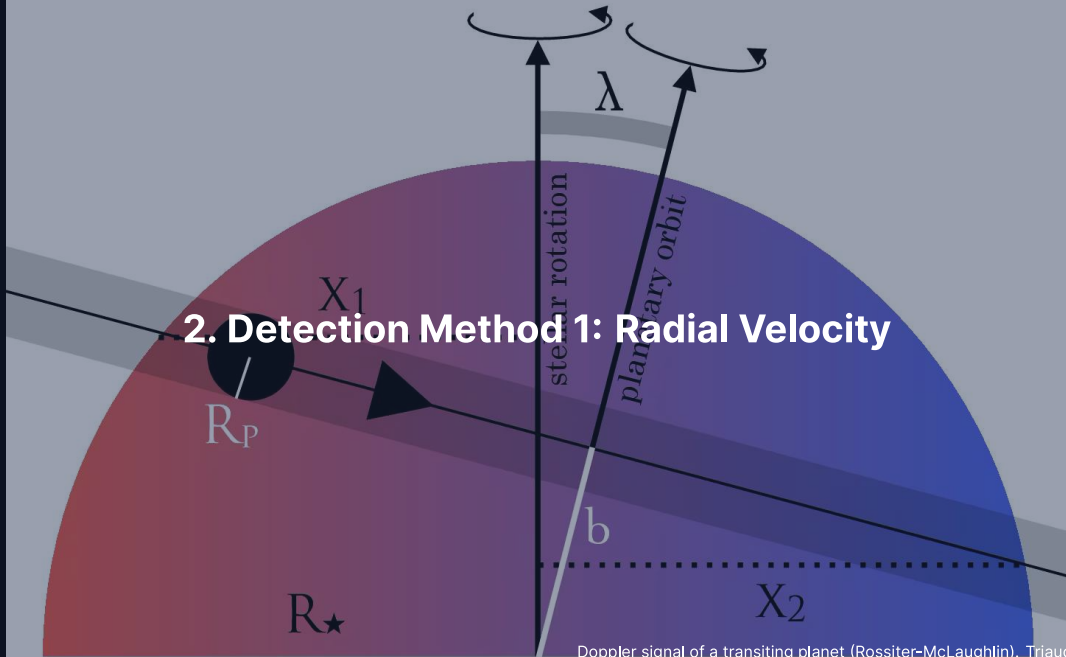
- **1992:** Wolszczan + Frail. Two planets around pulsar PSR B1257+12 from pulse-timing.
- **1995:** Mayor + Queloz. **51 Pegasi b** via radial velocity.  $0.5 M_J$ , 4.23-day orbit.
  - Contradicted every textbook picture of giant-planet formation.
  - → migration models reshaped formation theory.
  - Nobel Prize 2019.
- **1999:** HD 209458 b. First transit of a known RV planet.
  - Combined transit depth + RV → first bulk density of an exoplanet.
  - Confirmed hot Jupiters are gas-dominated.
- **2009–2018:** Kepler + K2 transformed the field statistically.
- **2022–present:** JWST atmospheric characterisation era.

## HD 209458 b: First Ground-Based Transit (1999)



- STARE photometer, two nights in Sep 1999.
- Successive transits dropped flux by  $\sim 1.7\%$ .
- Independent detection by Henry et al. (2000).
- After this point: exoplanets were physical objects, not just Doppler signals.

## 2. Detection Method 1: Radial Velocity



## Radial Velocity: Doppler Reflex of the Star

---

- Star + planet orbit a common barycentre (centre of mass).
- Star's reflex orbit:  $a_{\star} = (m_p / M_{\star}) a_p$ .
- Spectral lines blueshift / redshift periodically.
- For circular, edge-on orbit (semi-amplitude  $K_{\star}$ , the peak reflex velocity):

$$K_{\star} = \left( \frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

Quick scales:

- Jupiter at 5.2 AU:  $K_{\star} \approx 12.5 \text{ m s}^{-1}$ .
- Earth at 1 AU:  $K_{\star} \approx 9 \text{ cm s}^{-1}$ .



## Instrument Floor and Stellar Noise

---

- ELODIE (1995):  $\sim 10 \text{ m s}^{-1}$ .
- HARPS (2003):  $\sim 1 \text{ m s}^{-1}$ .
- ESPRESSO (2018):  $10 \text{ cm s}^{-1}$  demonstrated.
- Now **stellar noise**-limited: granulation, p-modes, starspots produce  $10 \text{ cm s}^{-1}$  to  $1 \text{ m s}^{-1}$  jitter even on quiet stars.
- True Earth twin: at the very edge of physical accessibility.

## The $m \sin i$ Degeneracy

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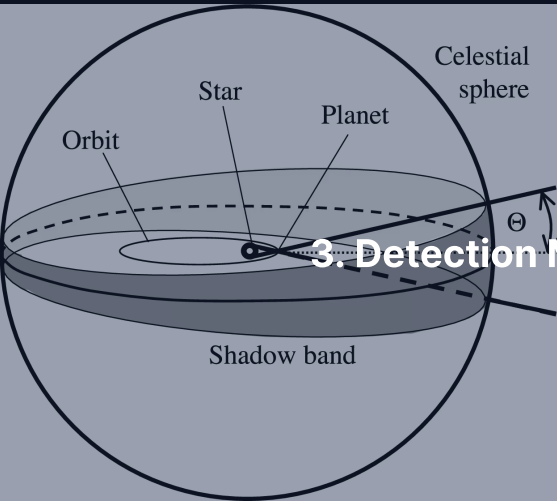
RV gives only  $K_*$ , hence  $m_p \sin i$ .

- Face-on:  $\sin i = 0$ , no signal at all.
- Edge-on:  $\sin i = 1$ , true mass.
- RV alone  $\rightarrow$  *minimum* mass only.

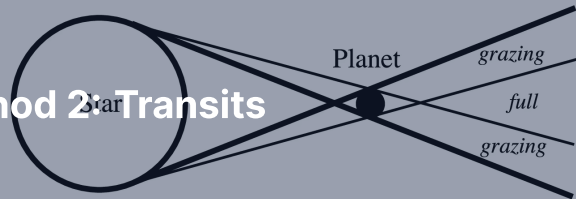
Break the degeneracy with:

- **Transit** ( $\sin i \approx 1$  to a few percent),
- **Astrometry** (gives inclination directly),
- **Direct imaging** (geometric).

RV biases: massive planets, short orbits, bright slowly-rotating quiet G/K stars.

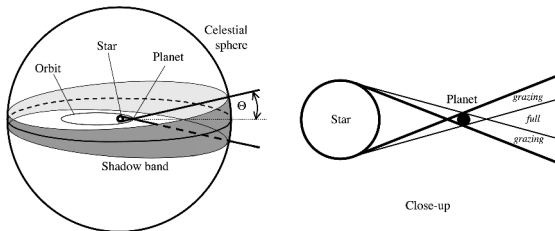


### 3. Detection Method 2: Transits



Close-up

# Transit Geometry



- Orbital plane near our line of sight  $\rightarrow$  planet crosses stellar disk.
- Depth:  $\delta = (R_p/R_\star)^2$ .
- Probability:  $\Theta \approx R_\star/a$ .
- Strong selection: short periods + large  $R_p/R_\star$ .

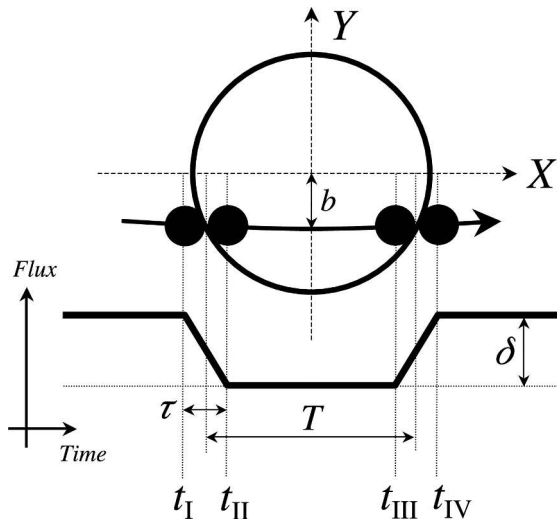
## Quick Scales

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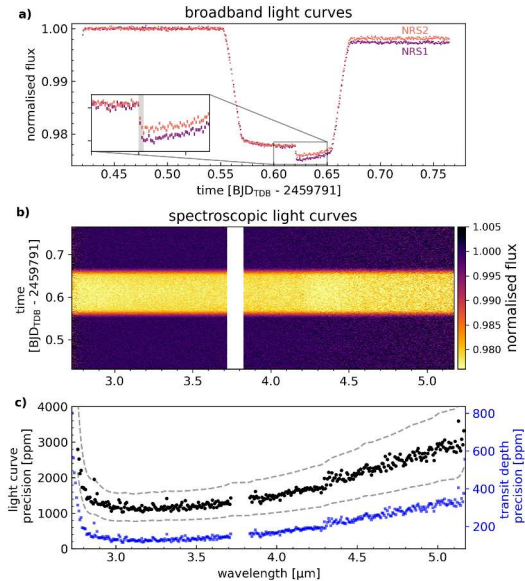
- Jupiter ( $R_J$ ) around Sun ( $R_\odot$ ):  $\delta \approx 1.01\%$ . Easy from the ground.
- Earth around Sun:  $\delta \approx 84$  ppm. Requires space.
- Earth around M dwarf (small, cool star;  $R_\star = 0.15 R_\odot$ ):  $\delta \approx 4000$  ppm. Ground-accessible!

Inverse scaling with stellar radius is why TRAPPIST-1 dominates atmospheric characterisation.

# Light Curve Anatomy



- Four contact times:  $t_I$ ,  $t_{II}$ ,  $t_{III}$ ,  $t_{IV}$ .
- Duration  $\rightarrow$  impact parameter  $b$  (transit offset from disk centre), inclination.
- Ingress shape  $\rightarrow$  stellar limb darkening.
- Depth  $\rightarrow$  planet radius (given  $R_*$ ).



JWST/NIRSpec transit photometry of WASP-39 b: transit depth measured to 200–600 ppm per spectroscopic bin. Alderson et al. (2023).

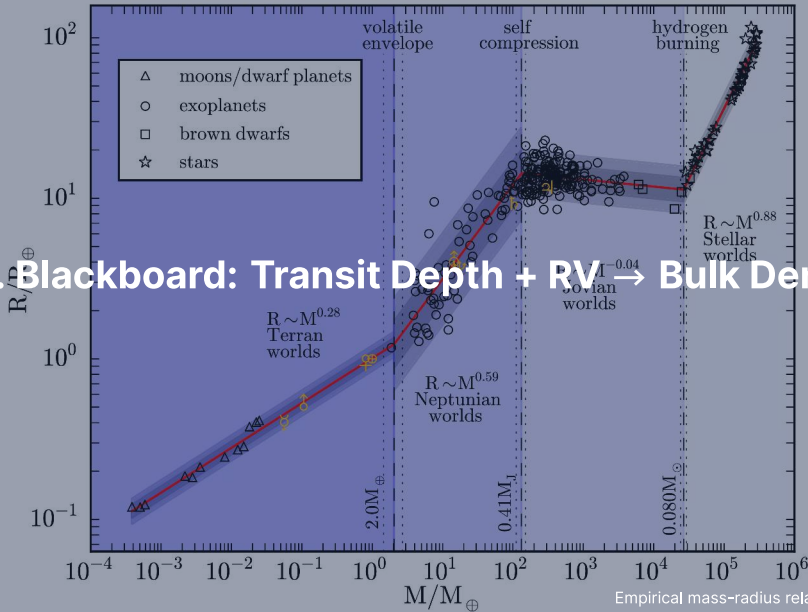
## Reading the JWST Transit Photometry

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- **Top:** raw broadband transit light curves, NIRSpec G395H NRS1 + NRS2.
- **Middle:** spectroscopic light curves at each wavelength bin.
- **Bottom:** per-channel precision  $\rightarrow$  200–500 ppm per integration.
- Photometric noise floor is now *below* typical exoplanet absorption features.
- Selection biases:
  - Short period ( $P < 100$  days) for repeated transits.
  - Large  $R_p/R_\star$  (favours hot Jupiters and M-dwarf systems).



# 4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



## Why Combine Transit and RV?

---

- Transit alone:  $R_p$ , but no mass.
- RV alone:  $m_p \sin i$ , but no radius.
- Combined: **both** for the same planet  $\rightarrow$  bulk density.
- Density is the central observable that turns a detection into a physical object.

## Step 1: Transit Depth

---

Star = uniform disk  $R_*$ . Planet = opaque disk  $R_p$ .

$$\frac{\Delta F}{F} = \frac{\pi R_p^2}{\pi R_*^2} = \left( \frac{R_p}{R_*} \right)^2$$

Given  $R_*$  from spectroscopy + parallax, the depth gives  $R_p$  directly.

Limb darkening + ingress shape  $\rightarrow \sim 10\%$  corrections.

## Step 2: RV Semi-Amplitude

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Conservation of momentum:  $M_{\star} a_{\star} = m_p a_p$ ,  $a = a_{\star} + a_p$ .

$$v_{\star} = \frac{2\pi a_{\star}}{P} = \frac{m_p}{M_{\star} + m_p} \cdot \frac{2\pi a}{P}$$

Use Kepler:  $a^3 = G(M_{\star} + m_p)P^2 / (4\pi^2)$ , sub in:

$$K_{\star} = v_{\star} \sin i = \left( \frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

$K_{\star} \propto P^{-1/3}$ : short-period planets give larger reflex.

## Step 3: Combine

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A transit forces  $\sin i \approx 1$  (orbit near edge-on).

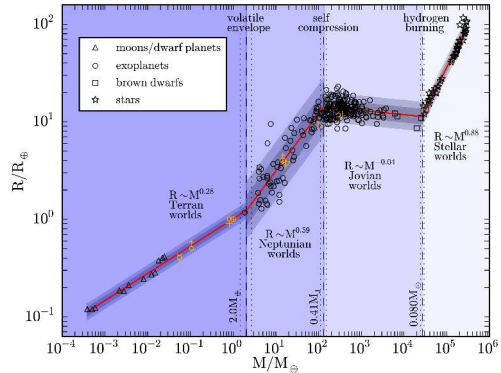
- $\sin i \approx 1$  collapses the  $m_p \sin i$  degeneracy.
- Both  $R_p$  and  $m_p$  measured for the same object.
- Bulk density follows from elementary geometry:

$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

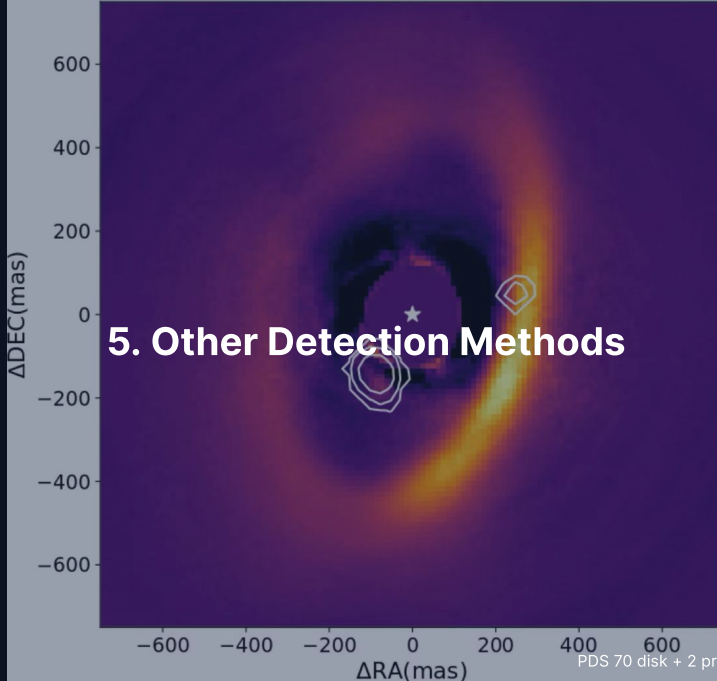
Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.

# Density Maps to Composition

- $\bar{\rho} \approx 5.5$ : rocky (Earth, Venus).
- $\bar{\rho} \approx 1.3$ : gas giant (Jupiter).
- $\bar{\rho} \approx 0.5$ : inflated H/He envelope.
- $\bar{\rho} \approx 2\text{--}4$ : *sub-Neptune*, the most common type with no solar-system analogue.



## 5. Other Detection Methods



PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

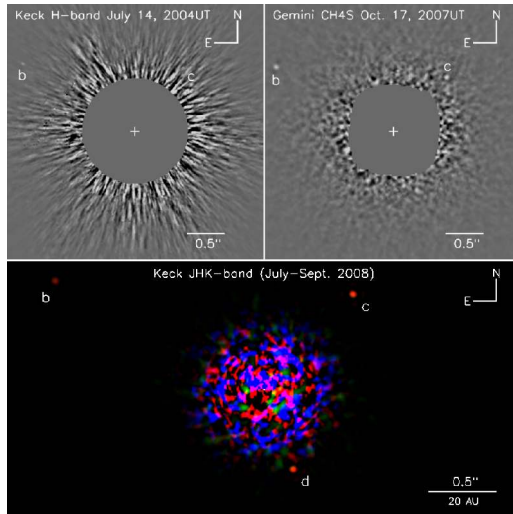
## Direct Imaging: Spatial Separation of Photons

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- Conceptually simplest, technically hardest.
- Sun + Jupiter contrast at visible:  $10^{-9}$ . Sun + Earth:  $10^{-10}$ .
- Four ingredients:
  - Adaptive optics (atmospheric turbulence correction)
  - Coronagraphs (stellar light suppression)
  - Angular + spectral differential imaging (speckle subtraction)
  - Long integration ( $\rightarrow$  averaging speckle realisations)
- Strongly biased to *young*, massive, wide-orbit, self-luminous giants.



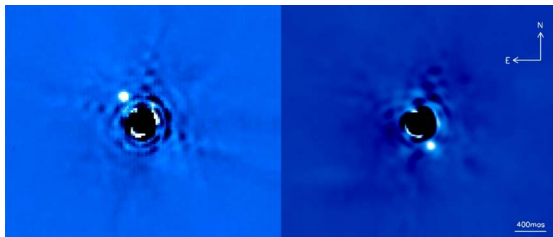
# HR 8799: Four Imaged Giants



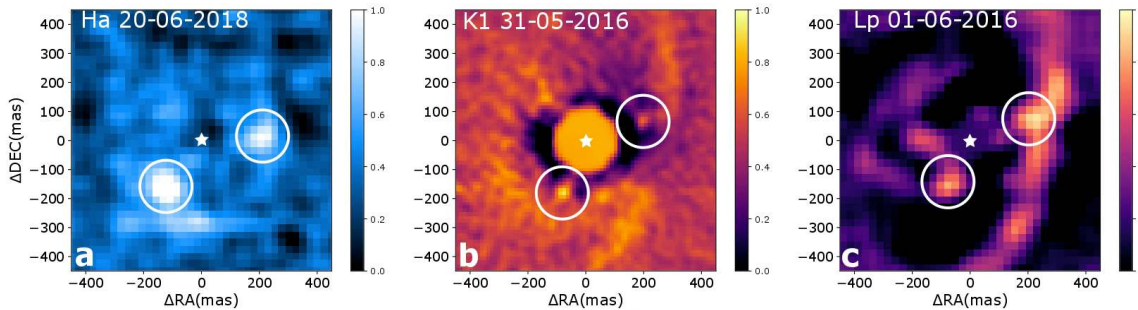
- Marois et al. 2008. Keck + Gemini AO; sub-panels show three discovery epochs (HK 2004, CH<sub>4</sub> 2007, JKH 2008) confirming bound co-moving companions.
- Three planets initially, fourth added 2010.
- Each  $\sim 5\text{--}10 M_J$  at 14–70 AU.
- Host: 30 Myr A-type. Planets still self-luminous and cooling.

## $\beta$ Pictoris b: Orbital Motion in Two Epochs

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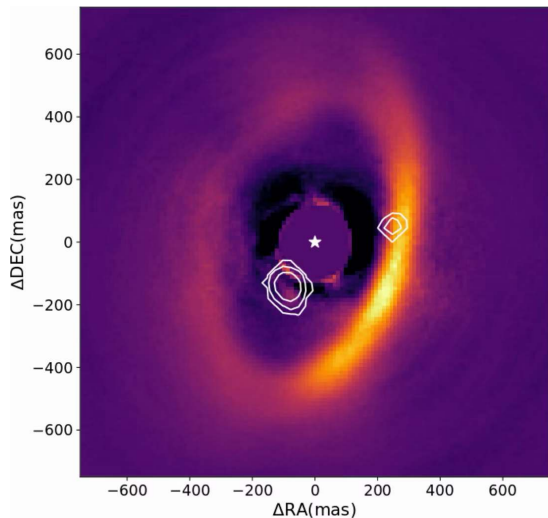


- Lagrange et al. (2010).
- $\sim 8 M_J$  at  $\sim 9$  AU.
- Two epochs (2003, 2009)  $\rightarrow$  orbital motion confirms bound companion.
- Same star: imaged debris disk (leftover dust ring) + a second giant (c).



PDS 70 b + c inside the disk gap in three bands: MUSE H $\alpha$  (a), SPHERE K1 (b), NACO L' (c).  
Haffert et al. (2019).

## PDS 70: Forming Planets in a Disk Gap



- Host: 5 Myr K7 with cleared central cavity.
- SPHERE (NIR) + MUSE ( $H\alpha$ ) detected b inside the gap (Keppler+ 2018).
- Second protoplanet c added later (Haffert+ 2019).
- Both are *actively accreting* ( $H\alpha$  shock emission).
- Cleanest direct match between planet-disk theory and observation.

## Astrometry: Gaia Will Find Jupiter Analogues

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$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- Jupiter analogue at 10 pc:  $\sim 0.5$  mas. Earth analogue:  $\sim 0.3 \mu\text{as}$ .
- Hipparcos (1989–93):  $\sim 1$  mas, no exoplanets.
- Gaia (2013–):  $20\text{--}50 \mu\text{as}$  per epoch,  $\sim 10 \mu\text{as}$  final.
- DR4 (late 2026):  $10^2\text{--}10^3$  astrometric exoplanets, mostly Jupiter analogues.
- Astrometry gives *inclination directly*  $\rightarrow$  true mass.

## Microlensing + Timing

---

- **Microlensing:** foreground star magnifies background source; planet adds a small spike.
  - Sensitive at kpc distances; sweet spot at 0.5–10 AU.
  - One-shot, never repeats.
  - Roman (2027): expected ~ 1000–3000 bound + comparable rogue planets.
- **TTVs:** gravitational coupling between transiting planets perturbs transit times → dynamical masses without RV.
  - Essential for faint stars: TRAPPIST-1 masses come almost entirely from TTVs.
- **Pulsar timing:** extraordinarily sensitive but tiny target sample.
- **Eclipsing binaries:** Kepler-16 b ("Tatooine") is circumbinary (orbits both stars).

## Detection Bias Summary

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- **RV**: massive, 1–10 yr orbits, bright G/K stars.
- **Transit**: short period, large  $R_p/R_*$ , quiet bright stars.
- **Direct imaging**: wide ( $> 10$  AU), young, massive, self-luminous.
- **Astrometry**: wide orbits matching mission baseline.
- **Microlensing**: 1–10 AU, any host distance, unrepeatable.
- **Timing**: compact multis (TTVs) or circumbinaries.

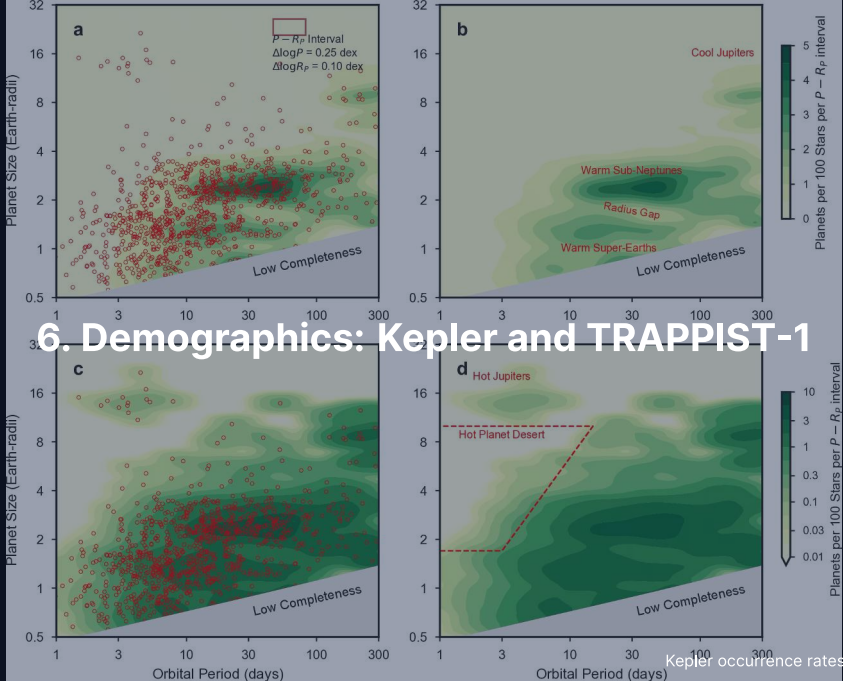
The shape of the observed exoplanet archive reflects the union of biases as much as the underlying physics.

# Break

End of Part 1: Detection Methods

Part 2: Demographics, Architectures, Atmospheres



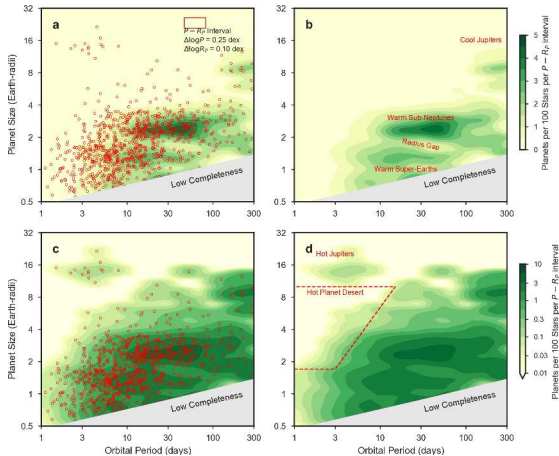


# The Kepler Revolution

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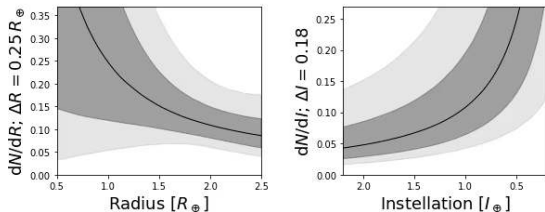
- 0.95 m telescope, single field, 150,000 stars, 4 years, 30 ppm precision.
- *Computable detection efficiency* → first occurrence-rate measurements (planets per star, bias-corrected).
- Headline: at least one planet per main-sequence star (Petigura+ 2018).
- Small planets ( $R < 4 R_{\oplus}$ ) dominate by far.
- Hot Jupiters: 0.5–1% of Sun-like stars (rare, but easy to find).
- $\eta_{\oplus}$  (Earth-size in HZ, the habitable zone):  $\sim 0.1$ – $0.6$ , central value  $\sim 0.3$  (Bryson+ 2021).

# Period-Size Occurrence Rate



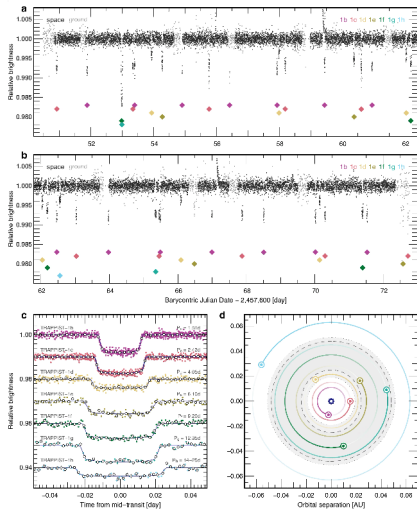
- California-Kepler Survey, after stellar parameter refinement (sub-panels label distinct period-radius regimes).
- Small planets common at every period.
- Slight fall-off at longer periods (partly observational).
- Long-period flattening → Kepler had only 4-yr baseline.

## $\eta_{\oplus}$ : Earths in Habitable Zones



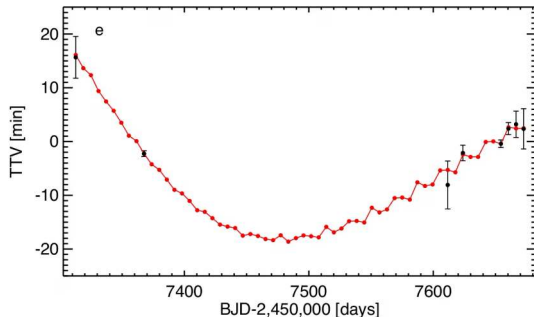
- Bryson+ 2021 marginal occurrence.
- Left: per unit radius. Right: per unit instellation (stellar flux received).
- Dark/light bands: 68% / 95% credible intervals.
- $\eta_{\oplus}$ : 0.1–0.6 depending on definition.
- **Earth-class planets are ordinary, not miraculous.**

# TRAPPIST-1: Seven Earths in Resonance



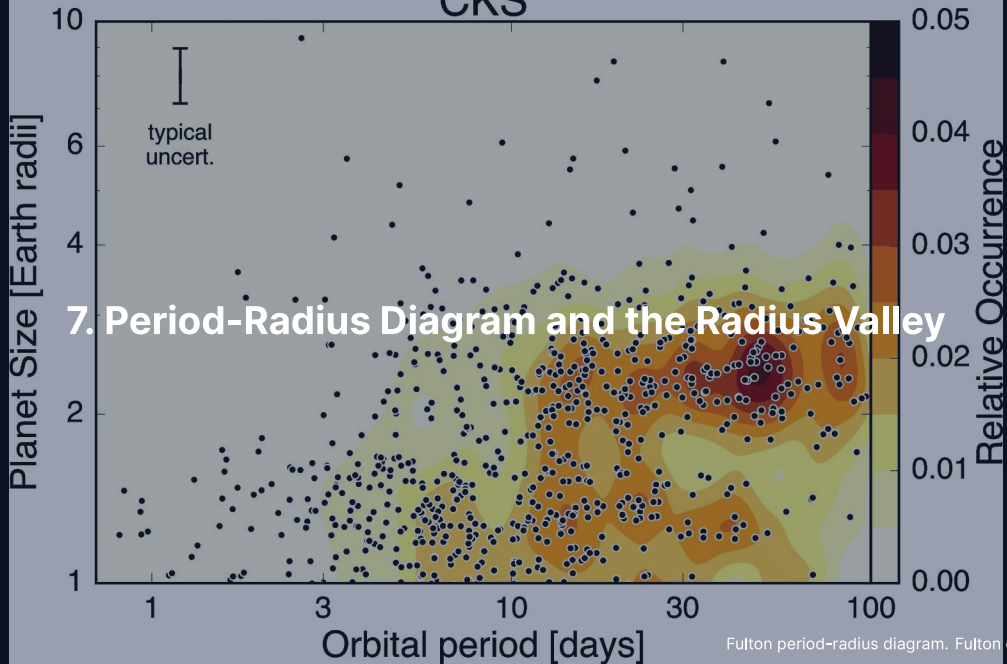
- M8 ultra-cool dwarf at 12 pc.
- Seven transiting Earth-sized planets within 0.06 AU.
- Five-step mean-motion resonant chain (period ratios near small integers) → early disk migration.
- Probability all seven transit if random:  $< 10^{-3}$ .
- Gillon et al. 2017.

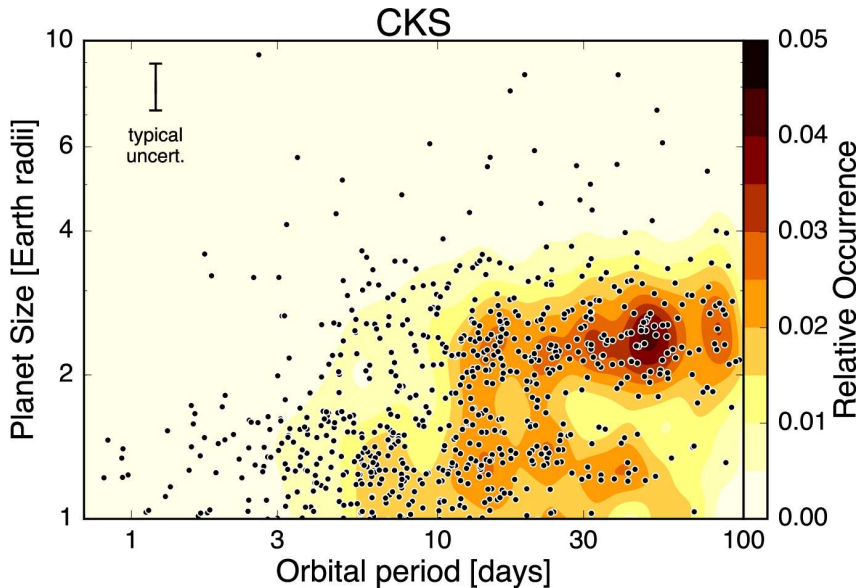
# TRAPPIST-1 Masses from TTVs



- TRAPPIST-1 e transit-timing variations.
- Tens-of-minutes amplitude over hundreds of days.
- Best-fit dynamical model includes the full 7-planet coupling.
- Yields all seven masses without RV.
- M8 is too faint for high-precision RV.

CKS





Period-radius distribution of small Kepler planets after stellar parameter refinement. Fulton et al. (2017).

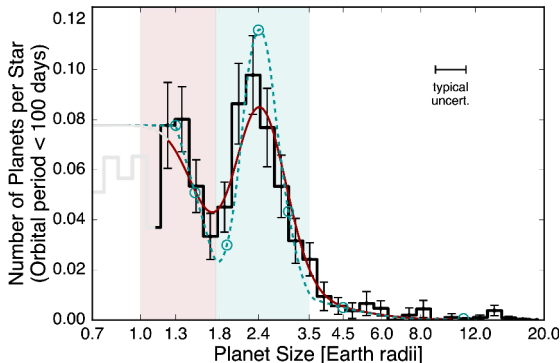


## Reading the Period-Radius Diagram

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- Hot Jupiters: tight clump at  $P < 10$  d,  $R > 10 R_{\oplus}$ .
- Sub-Neptunes ( $2\text{--}4 R_{\oplus}$ ): *the most common class*; no solar-system analogue.
- Super-Earths ( $1\text{--}1.8 R_{\oplus}$ ): rocky planets larger than Earth.
- Terrestrial regime ( $P > 100$  d,  $R < 1.5 R_{\oplus}$ ): largely unexplored. PLATO will fill.
- Most striking feature: the **deficit at**  $\sim 1.8 R_{\oplus}$ .

# The Radius Valley (Fulton Gap)



- Statistically significant deficit at  $1.5\text{--}2 R_{\oplus}$ .
- Below: super-Earths peaked at  $\sim 1.3 R_{\oplus}$ .
- Above: sub-Neptunes peaked at  $\sim 2.4 R_{\oplus}$ .
- Bimodal small-planet population. Defining demographic feature.

# Photoevaporation Strips H/He Envelopes

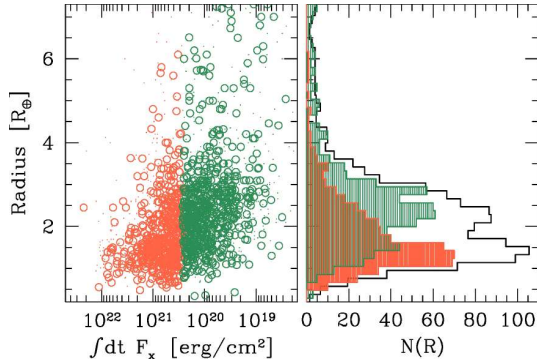
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Energy-limited escape rate:

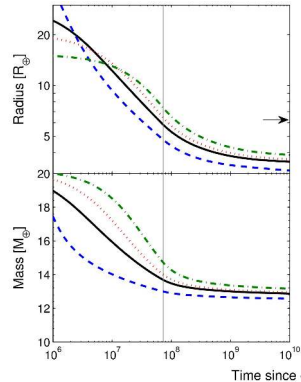
$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

- Young stars saturate XUV at  $\sim 10^3 \times$  present-day for  $\sim 100$  Myr.
- Low-density, low-mass planets are *easier to strip*.
- Cumulative loss for a  $10 M_{\oplus}$  sub-Neptune at 0.1 AU: a few % of an Earth mass.
- Comparable to a primordial H/He envelope on a rocky core.
- Owen & Wu (2013): naturally produces a bimodal radius distribution.

# Photoevaporation Model + Mass-Loss Tracks

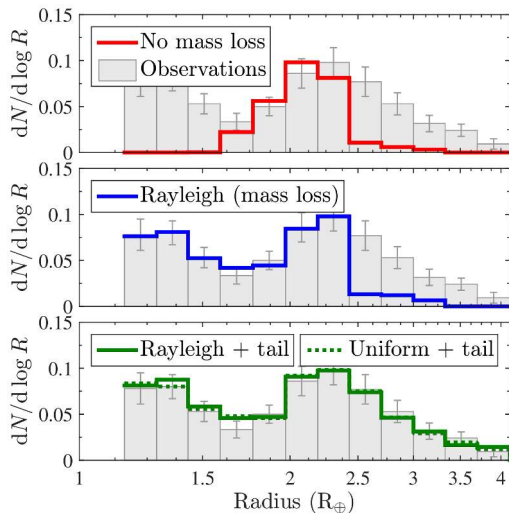


**Left:** predicted radius valley



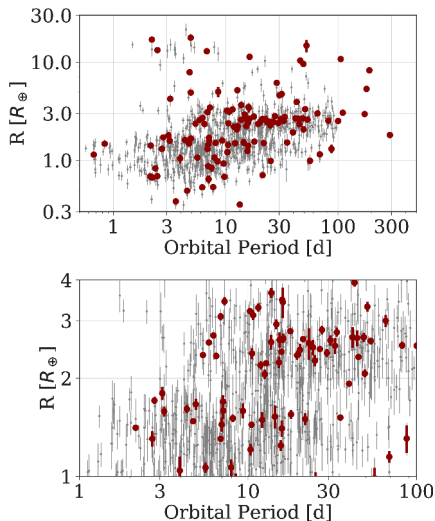
**Right:** time evolution of radius + mass; saturated XUV phase out to  $\sim 100$  Myr drives bimodal final distribution.

## Core-Powered Mass Loss: Same Outcome, Different Driver



- Ginzburg+ 2018, Gupta+ 2019.
- Hot rocky core retains formation heat  $\rightarrow$  basal flux into envelope drives slow outflow.
- No XUV needed.
- **Bottom panel** (Rayleigh + tail): bimodal radius distribution at  $\sim 1.8 R_{\oplus}$ , mirroring photoevaporation.
- Both mechanisms likely operate.

# Slope of the Valley with Period

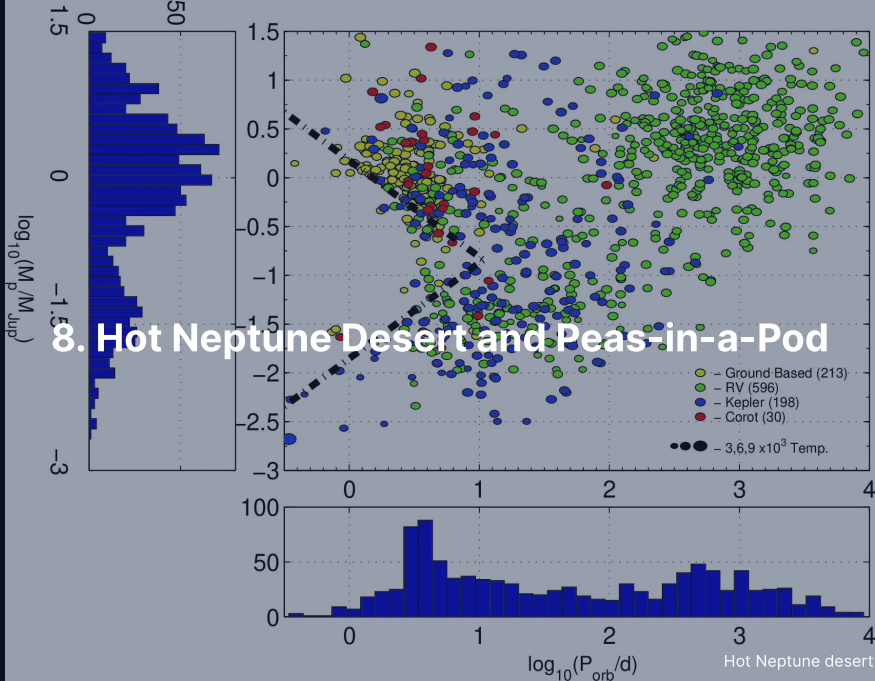


- Van Eylen+ 2018, asteroseismic radii (from stellar oscillations).
- Valley shifts to smaller  $R_p$  at longer periods.
- Brown: super-Earths. Blue: sub-Neptunes.
- Slope consistent with both photoevaporation and core-powered models.
- Current data cannot distinguish.

## Implication: Many Super-Earths Are Stripped Cores

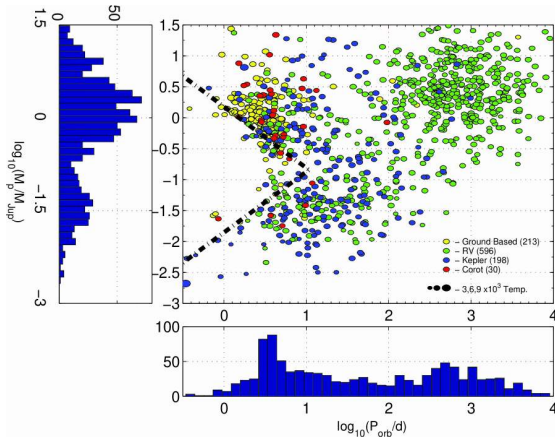
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- Many close-in rocky super-Earths are not “naturally” rocky.
- They are the bare cores of former sub-Neptunes.
- Whether a given super-Earth was born rocky or stripped: *cannot tell from bulk density alone*.
- Consequence for habitability: stripped cores have violent atmospheric histories and may have very different surface/interior chemistry.





# The Hot Neptune Desert



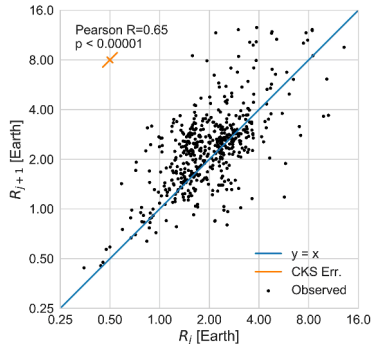
- Striking deficit at  $10\text{--}100 M_{\oplus}$ ,  $P < 5$  d.
- Hot Jupiters at the same periods: rare but present.
- Ultra-short rocky: rare but present.
- Hot Neptunes: *essentially absent*.
- Power-law boundaries (Mazeh+ 2016).

## Two Edges, Two Mechanisms

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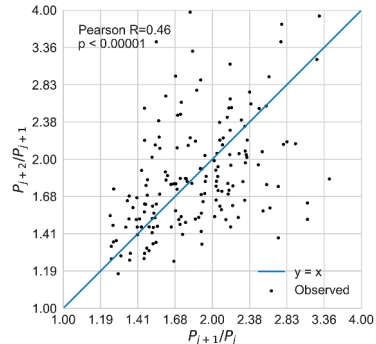
- **Lower edge (small mass):** photoevaporation strips weakly-bound Neptune envelopes → leaves super-Earths.
- **Upper edge (large mass):** Roche-lobe overflow (tidal gas stripping) on inflated hot Jupiters truncates the heavy population.
- Independent evidence that mass loss is the dominant sculptor of close-in planets.

# Peas-in-a-Pod



Adjacent radii: Pearson 0.65

Within a Kepler multi-system, planets tend to be *the same size* as each other and *uniformly spaced*. Weiss et al. (2018).



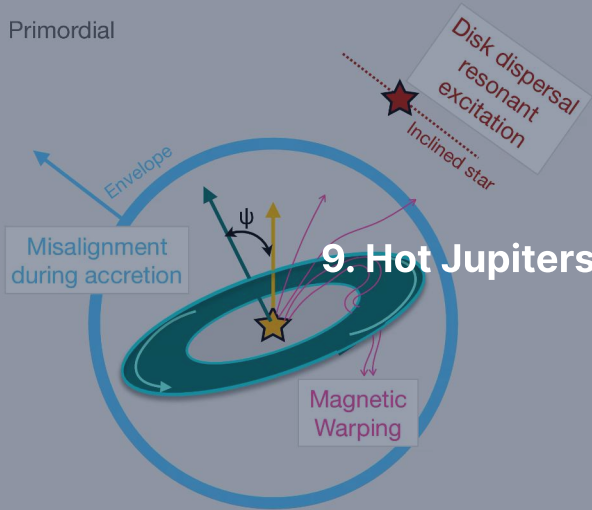
Adjacent period ratios: clustered

## Peas-in-a-Pod: Implications

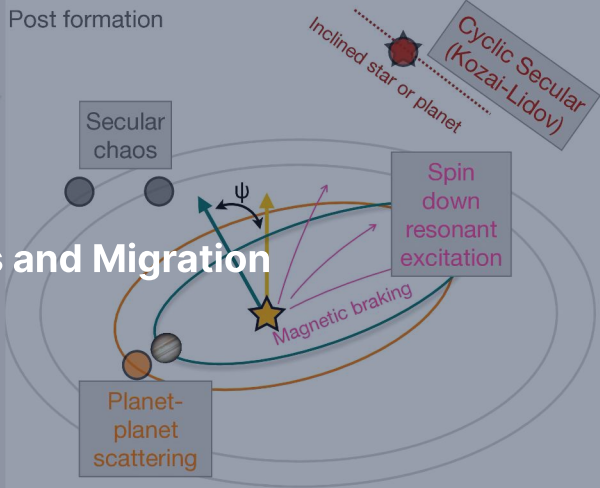
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- Implies smooth, local formation rather than stochastic giant impacts.
- Inner solar system (giant-impact dominated) may be unusual.
- TRAPPIST-1: extreme example of a peas-in-a-pod with resonant chain.
- Kepler-90 (8 planets), TOI-178 (6 planets in unbroken chain).
- Not universal: RV-discovered systems show much more architectural diversity.

Primordial



Post formation



## 9. Hot Jupiters and Migration

## Three Migration Channels

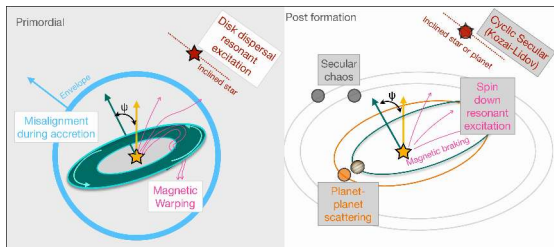
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Hot Jupiters at 0.05 AU *must* have migrated.

- **Disk migration** (Type II): smooth, gas-driven. Preserves alignment.
- **High-eccentricity migration** (Kozai-Lidov): distant perturber pumps eccentricity, tides circularise. Produces misalignments.
- **Planet-planet scattering**: dynamical instability ejects giants, leaves survivors on eccentric orbits, tides circularise. Misalignments.

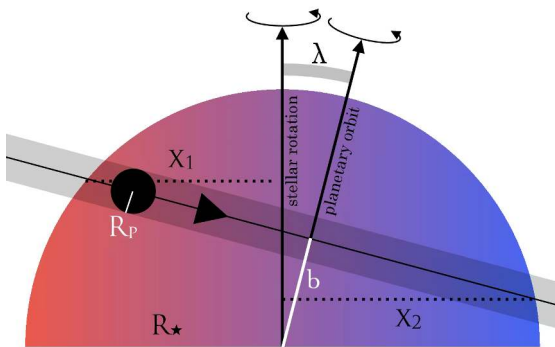
Inner edge set by Roche limit + tidal circularisation timescale: pile-up at  $\sim 0.04\text{--}0.05$  AU.

# Migration Channels and Stellar Obliquities



- Albrecht+ 2022.
- Disk migration  $\rightarrow$  aligned.
- Kozai-Lidov  $\rightarrow$  arbitrary obliquity.
- Scattering  $\rightarrow$  misaligned.
- Stellar obliquity (spin-orbit misalignment angle) is the main empirical handle.

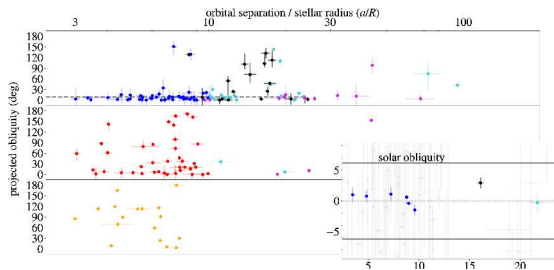
# Rossiter-McLaughlin Effect



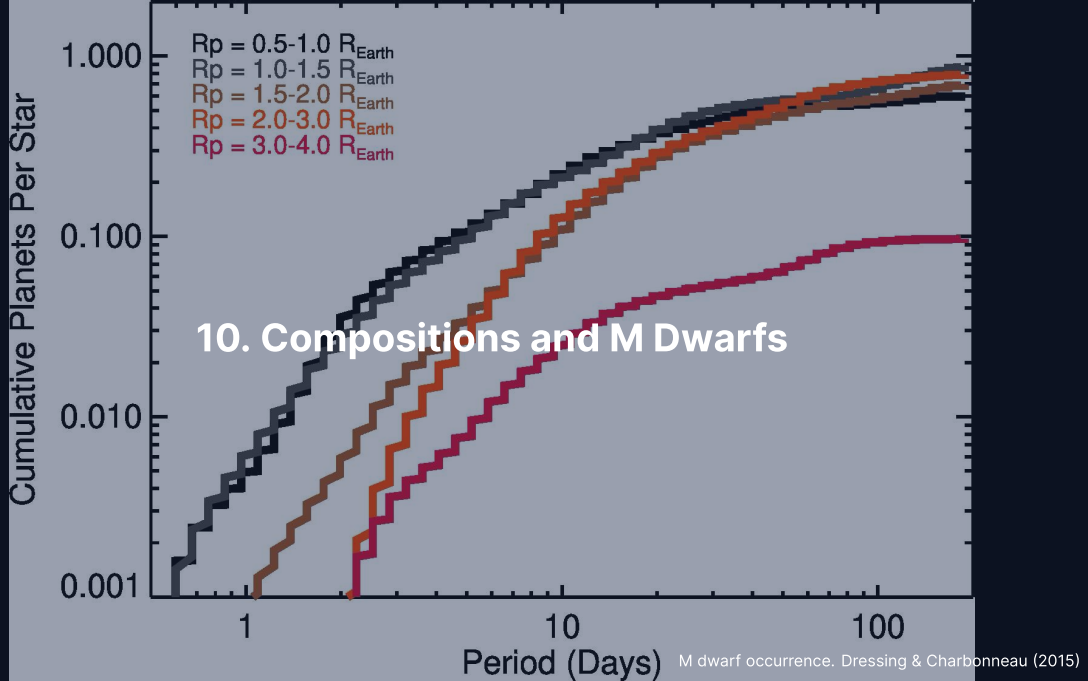
- Transiting planet first occults blueshifted (approaching) hemisphere, then redshifted.
- Time-resolved line distortion traces angle  $\lambda$  between projected stellar spin and orbit normal.
- Triaud (2018) review.



# Observed Obliquity Distribution



- $\lambda$  vs  $a/R_*$ , with sub-panels per host spectral type (panel labels too small for back rows; trust the trend, not the tick labels).
- Cool stars ( $T_{\text{eff}} < 6250$  K): tight orbits aligned (tidal realignment).
- Hot stars: scattered  $\lambda$ .
- Wide scatter at large  $a/R_*$  implies a wide *primordial* obliquity distribution.
- All three migration channels likely operate.

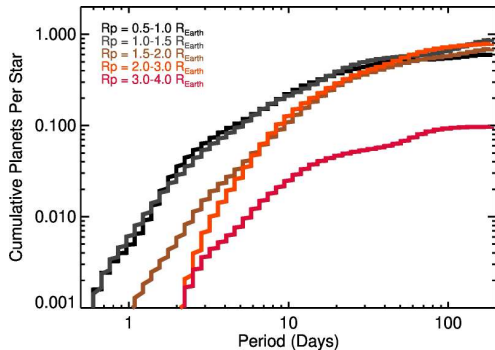


## Sub-Neptunes: Water World or H/He?

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- Super-Earths (below valley): rocky,  $\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$ .
- Sub-Neptunes (above valley):  $\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$ , need volatiles on top of a rocky core.
- Volatile component degenerate: thick H/He, thick H<sub>2</sub>O, or some mix.
- **Hycean** hypothesis (Madhusudhan+ 2023): shallow liquid ocean under thick H<sub>2</sub> atmosphere. Contested.
- Ultra-short-period rocky: thin atmosphere vs bare rock indistinguishable from density alone.

# M Dwarfs: Easy Targets, Tough Habitability

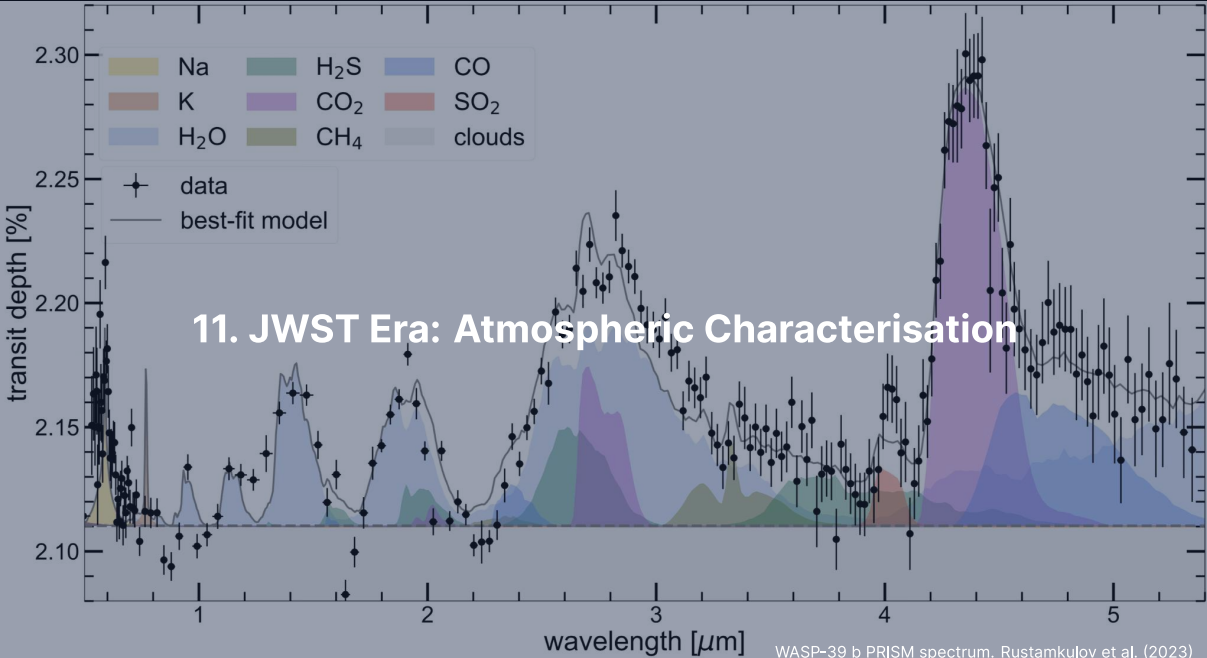


- ~ 75% of all main-sequence stars.
- M dwarfs host ~ 2.5 small planets each; ~ 0.16 in HZ.
- Transit + RV signals geometrically favourable.
- BUT: pre-main-sequence high-luminosity for hundreds of Myr.
- Habitable-zone planet *starts* inside runaway greenhouse zone.
- Dressing & Charbonneau (2015).

## M Dwarf Habitability: Two Worries

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- **Stellar activity** + early high-XUV → Luger & Barnes (2015) showed an ocean's worth of water can be lost as  $O_2$  (false-positive biosignature: a life-like signal, no biological cause).
- **Tidal locking**: HZ M dwarf planets in 1:1 spin-orbit resonance.
- Permanent dayside / nightside; climate must redistribute heat or condense volatiles on the night side.
- 3D GCMs (Yang+ 2013): substellar clouds can extend the inner edge.
- TRAPPIST-1 = central laboratory for these questions.



## Transmission Spectroscopy

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Starlight filtered through the atmosphere at transit encodes absorption.

Atmospheric scale height (density e-folding distance):

$$H = \frac{k_B T}{\mu m_u g}$$

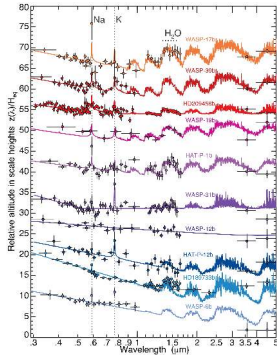
Wavelength-dependent depth:  $\delta(\lambda) = (R_p + n_H(\lambda) H)^2 / R_\star^2$ .

$H$  is wavelength-independent;  $n_H(\lambda)$  is the number of scale heights probed (typically a few).

Quick scales for hot Jupiter ( $T = 1500$  K,  $\mu = 2.3$ ,  $g = 25$  m s<sup>-2</sup>):

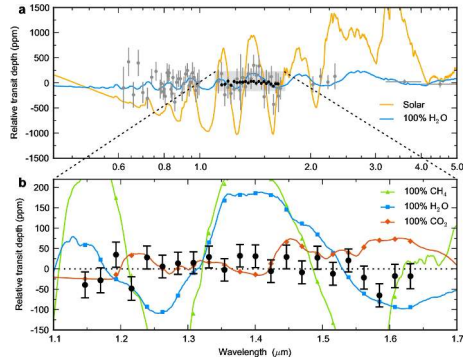
- $H \approx 200$  km
- Modulation across  $\sim 5$  scale heights:  $\Delta\delta/\delta \approx 2 \times 10^{-2}$
- Absolute modulation:  $\sim 200$  ppm  $\rightarrow$  JWST-routine.
- Sub-Neptune: 10–100 ppm. Earth/M dwarf:  $<10$  ppm.

# Clouds Can Mute Spectra



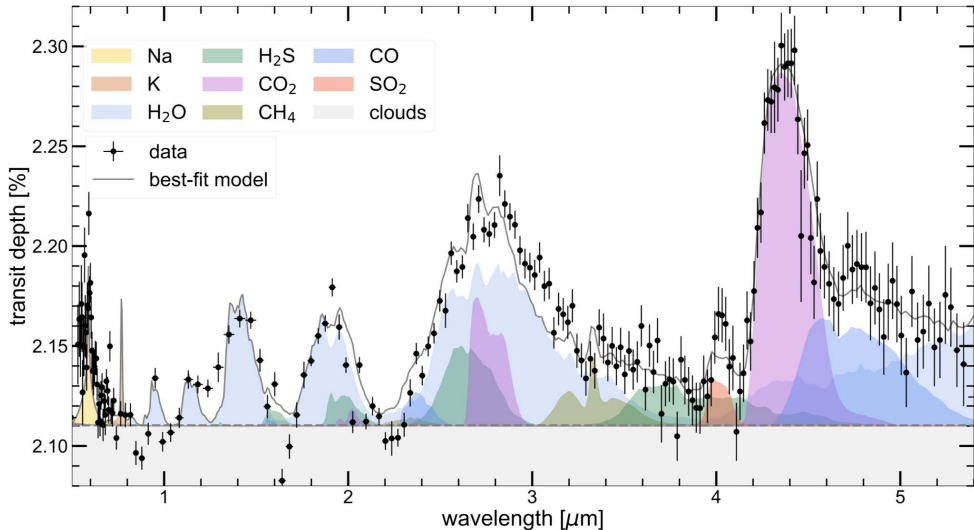
HST + Spitzer, ten hot Jupiters, ordered by optical-to-mid-IR altitude difference. Sing+ 2016.

Continuum from clear (Na, H<sub>2</sub>O at top) to aerosol-dominated (bottom).  
Diversity at similar  $T$ ,  $g$  is one of the central puzzles.



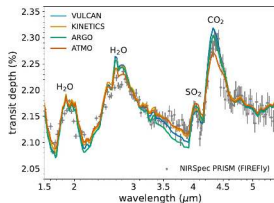
GJ 1214 b: featureless. Kreidberg+ 2014.



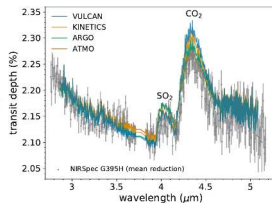


JWST/NIRSpec PRISM transmission spectrum of WASP-39 b. Shaded bands are model opacity contributions; the data support H<sub>2</sub>O (33 $\sigma$ ), CO<sub>2</sub> (28 $\sigma$ ), cloud (21 $\sigma$ ), Na (19 $\sigma$ ), CO (7 $\sigma$ ), not K, H<sub>2</sub>S or CH<sub>4</sub>. Rustamkulov+ 2023.

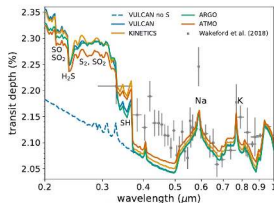
# WASP-39 b: First Disequilibrium Photochemistry



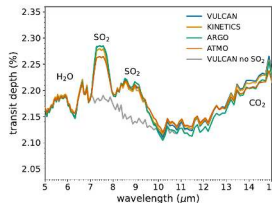
(a)



(b)



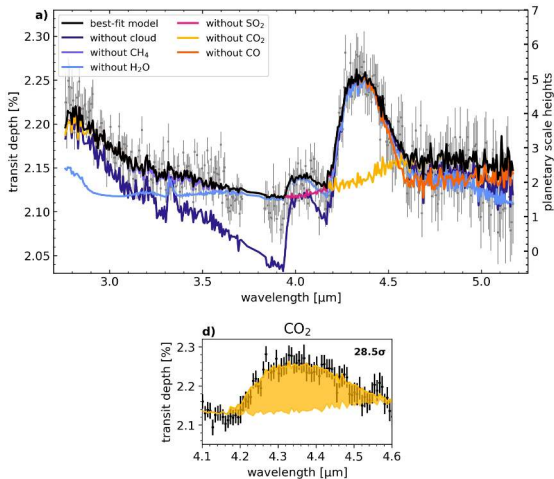
(c)



(d)

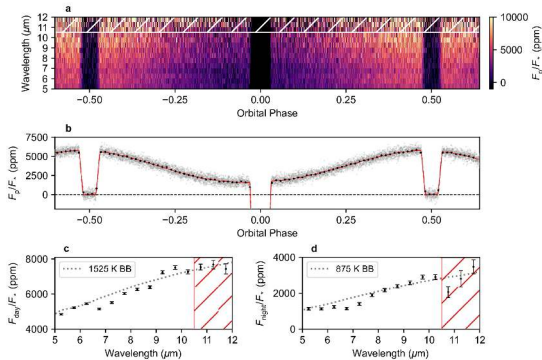
- ERS target:  $0.28 M_J$  hot Saturn.
- The  $4.05 \mu\text{m}$  feature is **photochemical  $\text{SO}_2$** .
- Panels are four wavelength ranges; each shows the same four codes (VULCAN, KINETICS, ARGO, ATMO).
- Panel d: MIRI prediction, grey is  $\text{SO}_2$  removed.
- $\text{SO}_2$  needs UV photolysis of  $\text{H}_2\text{S}$ , *not* equilibrium chemistry. Tsai+ 2023.

# WASP-39 b: $28.5\sigma$ CO<sub>2</sub> Feature



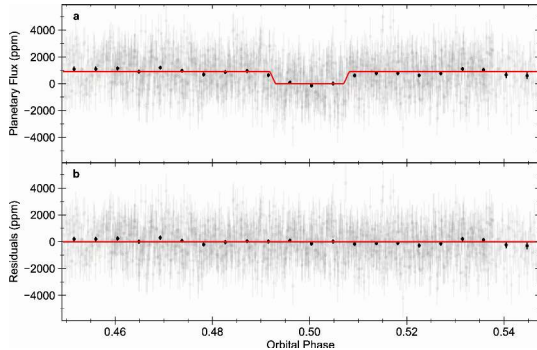
- Each coloured curve: best-fit model with one opacity removed.
- Panel a: full spectrum. Panel d: the CO<sub>2</sub> band alone.
- CO<sub>2</sub> near 4.3  $\mu\text{m}$ :  $28.5\sigma$ , against  $21.5\sigma$  (H<sub>2</sub>O) and  $4.8\sigma$  (SO<sub>2</sub>).
- Alderson+ 2023.

# WASP-43 b: A Phase Curve in Thermal IR



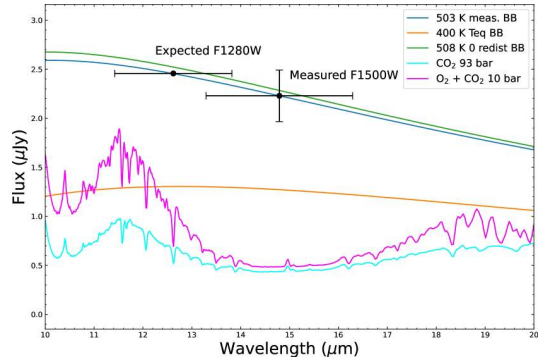
- JWST MIRI phase curve (brightness vs. orbital phase).
- Two transits, two secondary eclipses (planet passes behind the star), smooth phase variation.
- Strong day-night contrast despite thick atmosphere.
- Bell+ 2024.
- Day = hot, night = much colder.

# TRAPPIST-1 b: Bare Rock Dayside



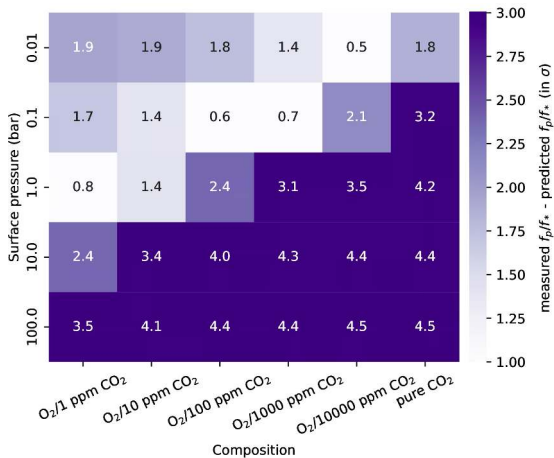
15  $\mu\text{m}$  secondary eclipse

$T_B = 503^{+26}_{-27}$  K, consistent with bare rock equilibrium. **Thick CO<sub>2</sub> atmosphere ruled out.** Greene+ 2023.



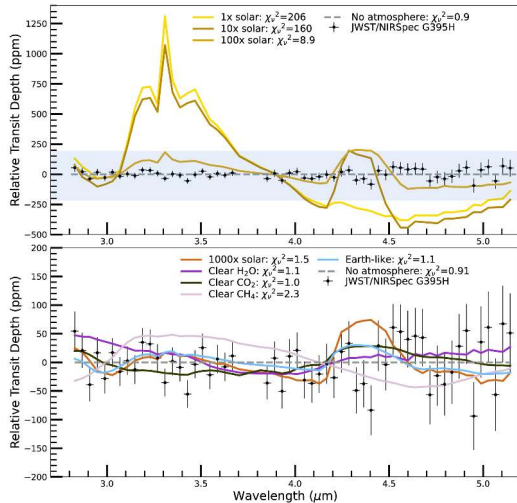
Compared to model atmospheres

## TRAPPIST-1 c: Same Story



- Zieba+ 2023.
- Grid of CO<sub>2</sub> partial pressure × thickness.
- $\geq 0.1$  bar of CO<sub>2</sub>: ruled out.
- Thin atmospheres or none: still possible.
- Inner TRAPPIST-1 planets do not retain Venus-like atmospheres.

# LHS 475 b: Featureless



- NIRSpec G395H, Earth-size at 12 pc.
- Flat featureless transmission spectrum.
- $\text{H}_2/\text{He}$  atmospheres ruled out.
- Pure  $\text{CH}_4$  ruled out.
- Pure  $\text{CO}_2$  Venus-like marginal.
- Lustig-Yaeger+ 2023.

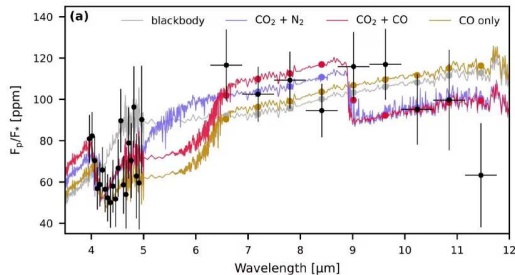
## Pattern: Most Small M Dwarf Planets Lack Atmospheres

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- LHS 475 b, GJ 486 b, GJ 1132 b, TRAPPIST-1 b/c: all consistent with no detectable atmosphere.
- Consistent with Luger & Barnes (2015): early M dwarf XUV strips primordial atmospheres.
- Maybe a watershed: M dwarf HZ planets may not retain water at all.

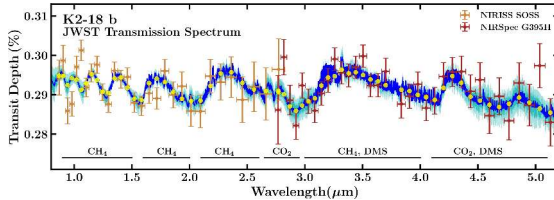


## 55 Cancri e: First Atmosphere on a Sun-Like Host?



- Hot rocky super-Earth,  $\sim 2000$  K dayside, plausibly molten.
- NIRCам + MIRI emission.
- Falls below bare-rock blackbody.
- Phase modulation  $\rightarrow$  secondary  $\text{CO}/\text{CO}_2$  atmosphere outgassed from molten surface.
- Hu+ 2024. *First* tentative atmosphere on rocky planet around Sun-like host.

## K2-18 b: Biosignature Controversy

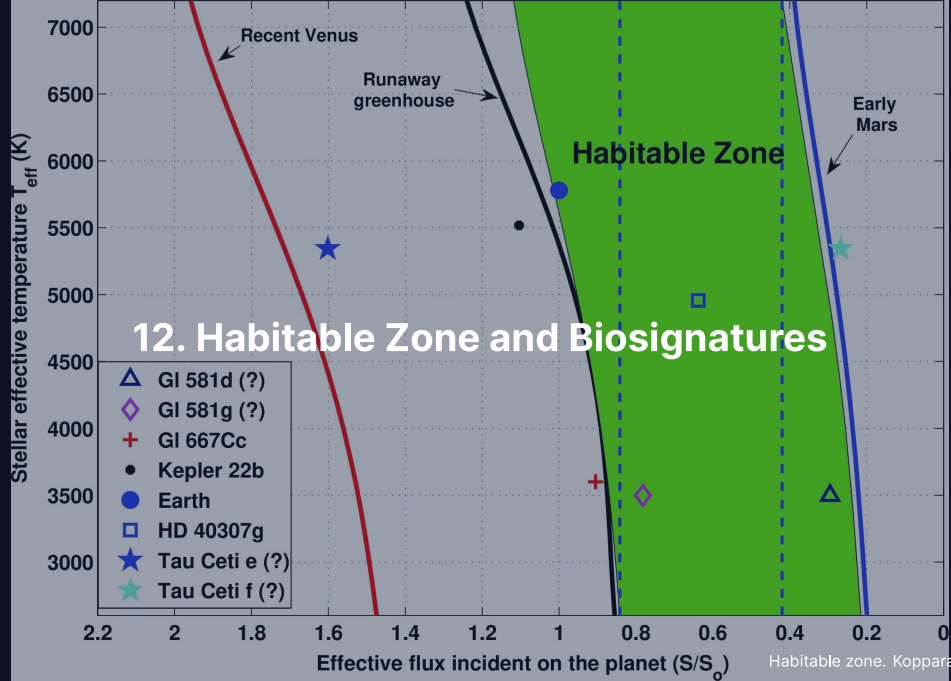


- Sub-Neptune in HZ of M3 host.
- Madhusudhan+ 2023:  $\text{CH}_4$  +  $\text{CO}_2$  + tentative DMS ( $\sim 3\sigma$ ).
- Interpreted as **hycean**:  $\text{H}_2$  atmosphere over ocean.
- Wogan+ 2024: cannot reproduce DMS; mini-Neptune fit.
- Glein+ 2024: geochemistry against habitable ocean.
- Retrieval-dependent. Debate open.

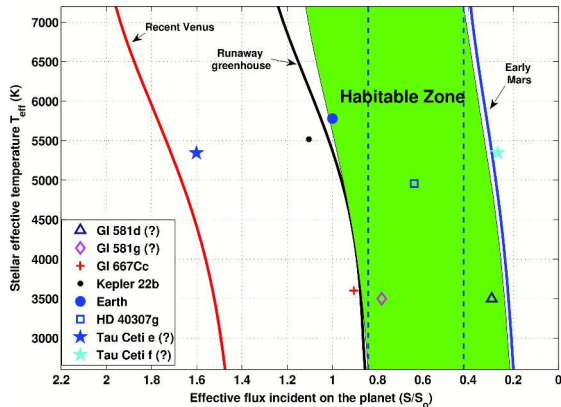
## K2-18 b's Pedagogical Value

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- Whatever the resolution: K2-18 b is the textbook case of how a tentative biosignature is tested in real time.
- Same pattern: ALH84001, Mars methane, Venus phosphine.
- This time the discussion has been faster, more transparent, better documented.
- Useful template for the louder claims of the next decade.



# The Classical Habitable Zone



- Inner edge: Simpson-Nakajima runaway greenhouse (L9).
- Outer edge: maximum  $\text{CO}_2$  greenhouse condenses out as ice.
- Kopparapu+ 2013 modernised Kasting+ 1993.
- Symbols: TRAPPIST-1, Tau Ceti, GJ 581/667, etc.

## HZ is a Trajectory, Not a Snapshot Line

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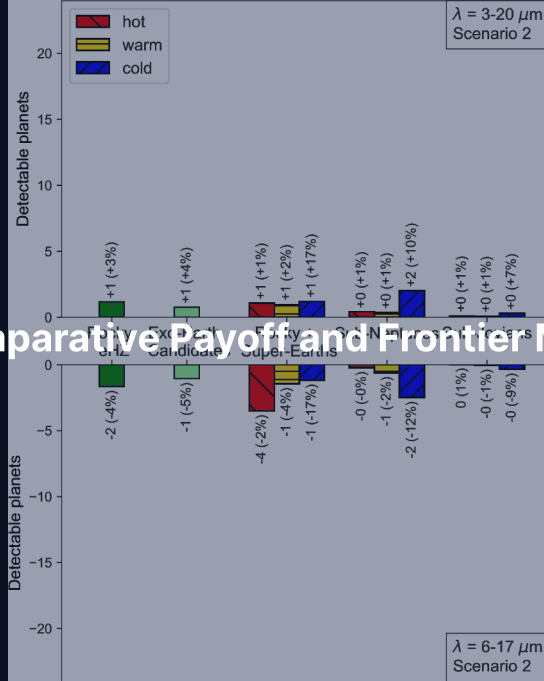
- History matters more than current location.
- Venus today sits formally inside the HZ if computed naively, yet is hellish (L9).
- M dwarf HZ planets traverse the runaway zone for hundreds of Myr in pre-MS phase.
- 1D Kasting/Kopparapu boundaries shift by 5–20% in 3D GCMs.
- Treat the HZ as a first-order screening tool, not a precise boundary.

# Biosignature Gases and False Positives

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- Classical: *disequilibrium combinations*. On Earth:  $O_2 + CH_4$ .
- Catalogue:  $O_2$ ,  $O_3$ ,  $CH_4$ ,  $N_2O$ , organosulphur, organohalogen species.
- No single gas in isolation is convincing.
- **False positives are the main challenge.**
  - Wordsworth+ 2014:  $H_2O$  photolysis + H escape  $\rightarrow$  abiotic  $O_2$ .
  - $CO_2$  photolysis  $\rightarrow$  abiotic  $O_2$  via  $CO + O$ .
  - Volcanic + serpentinisation  $\rightarrow$  abiotic  $CH_4$ .
- Inverse problem: must rule out *all* plausible abiotic pathways.

# 13. Comparative Payoff and Frontier Missions





## Is the Solar System Typical?

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Honest answer: **not obviously, but not yet known to be rare.**

Sparsely populated corners:

- Most common host: M dwarf, not G dwarf.
- Most common planet: sub-Neptune ( $2\text{--}3 R_{\oplus}$ ). *Solar system has none.*
- Most common architecture: peas-in-a-pod,  $\sim 5\text{--}8$  similar planets within 0.2 AU. *Not the inner solar system.*
- No hot Jupiter or hot Neptune in our system.
- Wide low-eccentricity giants (vs RV-discovered eccentric giants).

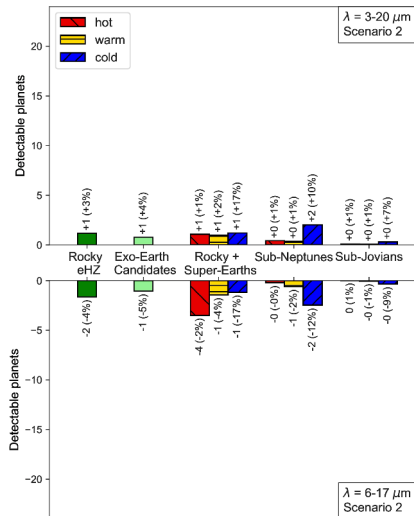
But: detection biases penalise solar-system analogues. Gaia DR4/DR5 + PLATO will resolve.

## Frontier Missions 2026–2035: Transits and Atmospheres

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- **PLATO** (ESA, end 2026): Earth analogues in HZ of bright Sun-like stars; asteroseismic stellar radii.
- **Ariel** (ESA, 2029): atmospheric statistics of ~ 1000 exoplanets at 0.5–7.8  $\mu\text{m}$ .
- **Roman** (NASA, 2027): 1000–3000 microlensing planets at 1–10 AU + comparable rogue population; coronagraph tech demo at  $10^{-8}$ – $10^{-9}$ .

# Frontier Missions 2030s–2040s: Direct Imaging



- **HWO** (NASA, 2040s): 6 m,  $10^{-10}$  contrast, ~ 25 Earth analogues.
- **LIFE** (ESA concept): nulling interferometer (starlight cancelled by interference), 4–18  $\mu\text{m}$  thermal emission.
- Figure: yield-sensitivity panels from Quanz+ 2022; trends matter, not the small numerical labels.
- **ELT class** (ground, 2028+): 39, 24.5, 30 m. High-contrast + high-resolution NIR.

# From Demography to Individual Characterisation

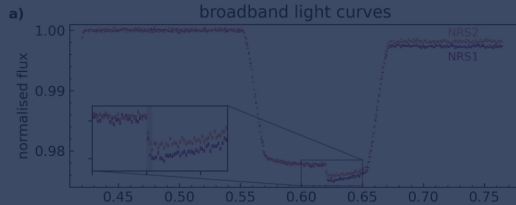
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- Today: thousands of detected planets, dozens of well-characterised atmospheres.
- 2040s: full atmospheric retrievals on a curated sample of Earth analogues.
- Open question for L14: what would convince us we have *detected life*?
  - Single biosignature gas?
  - Specific combinations at specific abundances?
  - Temporal variability (seasonal cycles)?
  - Vegetation or photosynthetic surface features?

## Summary: Today's Course-Level Takeaways

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1. Each detection method picks out a different region of parameter space; the observed archive is the union of biases.
2. Transit + RV  $\rightarrow$  bulk density: the observational machinery that turned exoplanet detections into physical objects.
3. Kepler showed planets are common.  $\eta_{\oplus} \sim 0.3$ . Hot Jupiters are rare.
4. **Radius valley** at  $\sim 1.8 R_{\oplus}$ : photoevaporation + core-powered mass loss as universal sculptor.
5. Hot Neptune desert + peas-in-a-pod + TRAPPIST-1 chain: three architectural results that any formation theory must explain.
6. JWST: WASP-39 b  $\text{SO}_2$ , TRAPPIST-1 b/c bare rock, K2-18 b biosignature controversy.
7. Solar system is not obviously typical. Gaia DR4/5 + PLATO will tell us if it is rare.



# Questions?

Next: **Lecture 14. Synthesis & Astrobiology**

Lecture notes: [formingworlds.github.io/IntroductionToPlanetaryScience](https://formingworlds.github.io/IntroductionToPlanetaryScience)

